

28	B	A

9749/01/ASRJC/2023Prelim

[Turn Over

	$P = \frac{E}{t} = \frac{n_p E_p}{t}$ $\frac{n_p}{t} = \frac{P}{E_p} = \frac{P\lambda}{hc}$ $\frac{n_e}{t} = \frac{1.5}{100} \left(\frac{n_p}{t} \right) = \frac{1.5}{100} \left(\frac{P\lambda}{hc} \right)$ $I = \frac{n_e e}{t} = \frac{1.5}{100} \left(\frac{P\lambda}{hc} \right) e$ $= \frac{1.5}{100} \left(\frac{0.080(546 \times 10^{-9})}{hc} \right) e$ $= 5.27 \times 10^{-4} \text{ A} = 0.53 \text{ mA}$	
20	C	E